

Dataset Selection and Feature Extraction for the Unified Performance Index (UPI)

This project leverages StatsBomb Open Data, an industry gold standard that makes professional, elite-level football datasets publicly available via their GitHub repository. Specifically, we are extracting data from matches that include 360-degree tracking data, such as the 2018 FIFA World Cup and the Lionel Messi Data Biography. The dataset contains a large number of records, with over 200,000 events in total.

Feature Extraction: Strategy Raw football event datasets are highly sparse, often containing over 100 columns where many are irrelevant to non-terminal actions.

To build our Markovian State-Value framework and Event-Level Expected Possession Value (EPV) model, we must aggressively filter out the noise. The core variables we extract are:

- **location (x,y) and pass_end_location:** These features represent the ball's location before and after each action, allowing it to be mapped to specific states in a Markov model, track spatial progression across pitch zones, and ultimately compute Expected Threat (xT) by assessing how positional changes affect the probability of scoring.
- **under_pressure:** A boolean indicating defensive pressing, which differentiates between safe states and dangerous, high-pressure states. It identifies whether an action occurs while the player is being actively challenged by opponents. It adds contextual information about the surrounding defensive situation, enabling the model to reflect the level of difficulty associated with each action.
- **shot_statsbomb_xg:** This is a pre-calculated probability that a shot results in a goal. It serves as an anchor for terminal states in the Markov model, ensuring that scoring actions are grounded in realistic probabilities.
- **360_frames:** The 360° freeze-frame data records the positions of all visible players at each event. It represents the full game context, including team shape and player positioning, and supports advanced spatial analysis such as pitch control.

Visualizations and Purpose: To accurately value a player's decision-making (and solve the outcome bias known as the "FIFA Paradox"), spatial coordinates alone are insufficient.

Visualization 1: Raw Event Data (Spatial Locations): As seen in this traditional pitch map, extracting just the location and pass_end_location shows us what happened. However, it completely ignores the defensive context, failing to capture the difficulty of the passes

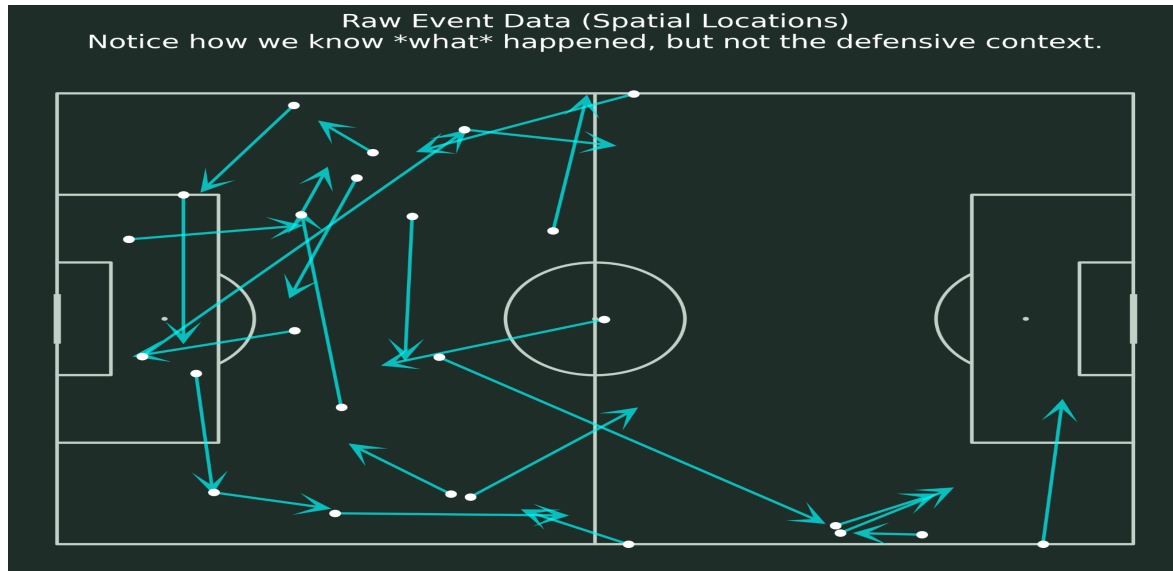


Diagram:1 Spatial_event_map

Visualization 2: StatsBomb 360 Data (Defensive Pressure & Pitch Control): By extracting the 360_frames and under_pressure features, we can map the exact positioning of teammates and opponents.

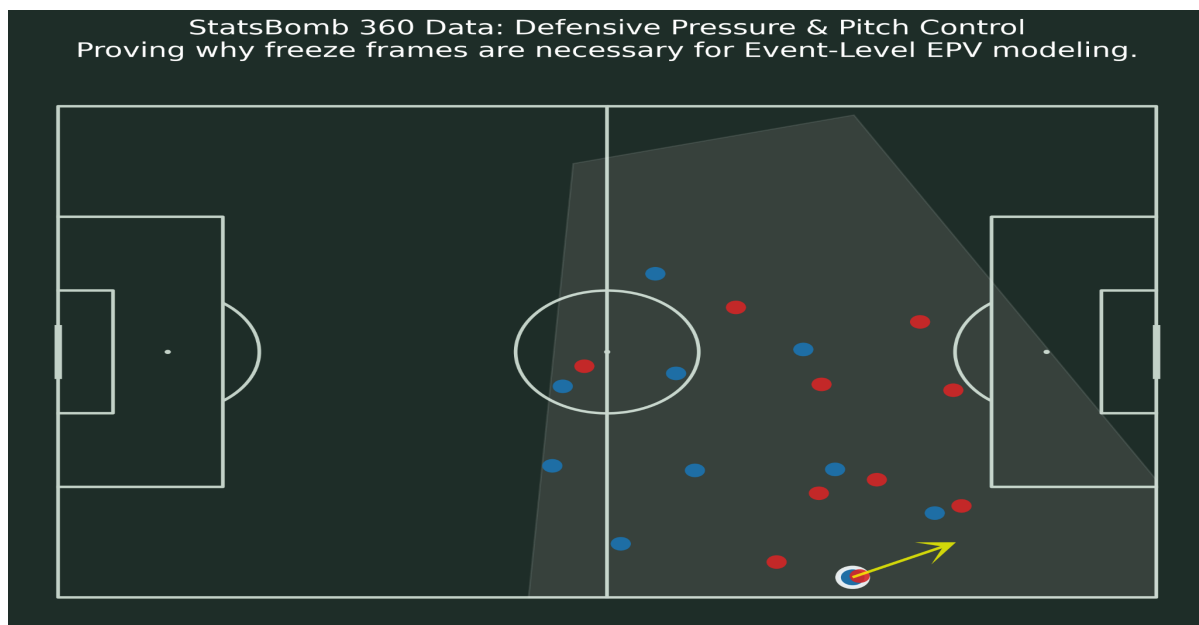


Diagram:2 360_freeze_frame

The purpose of using this specific tracking data is to calculate "Pitch Control" (Event-Level EPV). Visualizing the defense's structural shape proves why these features are strictly necessary: they allow the model to mathematically reward defensive midfielders and playmakers for executing actions that successfully bypass opponent pressure, quantifying the "invisible" tactical contributions that traditional stats ignore.

